

DEEP LEARNING–BASED NEURAL DECODER FOR EEG MOTOR IMAGERY

Phase 2 — Methodology and Implementation Plan

1st Review • 15 September 2026

Scope of this document

This document covers the seven Phase 2 deliverables: proposed methodology, system architecture, model and algorithm selection, dataset selection and description, tools and environment, implementation plan with milestones, and the experimental plan with evaluation metrics. It builds directly on the literature survey submitted at the 0th Review.

1. Problem Recap and Research Gap

Electroencephalography records the brain's electrical activity non-invasively, and motor imagery — imagining a movement without performing it — produces characteristic changes in the mu (8–13 Hz) and beta (13–30 Hz) rhythms over the sensorimotor cortex. A decoder that maps these signals to intended movements would let people with severe motor impairment control computers, wheelchairs and prosthetic limbs without moving.

The obstacle is not classification accuracy in the abstract; it is that accuracy collapses when the model meets a person it was not trained on. EEG is shaped by individual skull thickness, cortical folding, electrode placement and the strategy a person adopts when imagining movement, so the feature distribution shifts from subject to subject. Our literature survey quantified this directly.

Model	Within-subject accuracy (IV-2a)	Cross-subject accuracy (IV-2a)	Drop
CTNet (2024)	82.52%	58.64%	–23.88 pts
CTNet (2024), IV-2b	88.49%	76.27%	–12.22 pts

Figure/Table 1.1 — The generalisation penalty reported by CTNet. The same architecture loses nearly 24 accuracy points on the 4-class task when the test subject is unseen during training. Source: Zhao et al., *Scientific Reports* 14:20237 (2024).

A model that scores 82% on people it has met and 59% on a stranger is not deployable. A patient cannot be asked to supply hours of labelled calibration data before their wheelchair responds. This gap — high accuracy on unseen subjects, achieved with a model small enough to run on assistive hardware, and with little or no calibration — is the problem this project targets.

Research gap (carried forward from the 0th Review)

Existing motor imagery decoders achieve strong within-subject accuracy but degrade sharply on unseen subjects. Recent architectures close this gap by adding capacity — attention layers, transformer depth, larger parameter counts — which conflicts with the low-power, low-latency requirements of real assistive devices. It is not established whether the improvement in cross-subject decoding comes from architectural capacity or from making the input representation subject-invariant in the first place.

2. Research Hypothesis and Objectives

The 0th Review identified the gap but did not commit to a testable claim. Phase 2 closes that. The project is organised around one primary hypothesis and two secondary hypotheses, each falsifiable by a specific experiment.

2.1 Primary hypothesis

H1

For cross-subject motor imagery decoding, making the input representation subject-invariant through covariance alignment contributes more to leave-one-subject-out accuracy than an equivalent increase in model capacity, and its benefit is largely complementary to feature-space domain-generalisation losses rather than redundant with them.

Tested by: the ablation grid in Section 10. If alignment and added capacity contribute equally, or if alignment's benefit disappears once domain-generalisation losses are present, H1 is falsified — and that is itself a reportable finding, because it would tell the field that architectural scaling is the productive direction.

2.2 Secondary hypotheses

- **H2.** Windowed self-attention matches or exceeds full self-attention under leave-one-subject-out evaluation while using materially fewer parameters, because unconstrained attention capacity overfits subject-specific structure on datasets of this size (~2,600 trials).
- **H3.** Subject-invariant alignment reduces the number of labelled target-subject trials needed to reach a fixed accuracy threshold, measured as a calibration-efficiency curve.

2.3 Objectives

1. Build a reproducible preprocessing and evaluation pipeline for motor imagery EEG with a single, fixed leave-one-subject-out protocol applied identically to every model.
2. Reproduce four published architectures — EEGNet, ATCNet, CTNet and EEG Conformer — under that one protocol, so their cross-subject performance can be compared on equal terms for the first time.
3. Design and implement SAND-Net, a lightweight hybrid decoder that combines one justified component from each reviewed architecture with an explicit subject-invariance mechanism.
4. Run a component-wise ablation that isolates the contribution of alignment, multi-scale temporal modelling, windowed attention and the domain-generalisation losses.
5. Quantify calibration efficiency: accuracy as a function of the number of labelled trials available from the unseen subject.
6. Validate model interpretability against known neurophysiology — specifically, whether learned channel importance reproduces contralateral event-related desynchronisation.

Why objective 2 matters on its own

The four papers report accuracies obtained under different train/test splits — hold-out, 5-fold, session-based, subject-specific. The published numbers therefore cannot be ranked against each other. Evaluating all four under one identical protocol is a contribution independent of whether our own model wins.

3. Proposed Methodology

The system is a five-stage pipeline. Stages 1 and 2 are signal processing, stage 3 is the learned decoder, stage 4 is the training objective, and stage 5 is interpretation.

Stage	Operation	Purpose
1. Acquisition	Load raw EEG epochs, select channels, reject bad trials	Standardised input across three datasets
2. Signal conditioning	Band-pass filter → epoch → Euclidean Alignment → normalise	Remove noise; remove subject-specific covariance shift
3. Feature encoding	Depthwise-separable convolution → multi-scale temporal branch → windowed self-attention	Learn spatial filters, multi-band temporal dynamics, long-range dependencies
4. Objective	Cross-entropy + supervised contrastive + domain-adversarial loss	Force class structure to be shared across subjects
5. Interpretation	Class Activation Topography + attention heatmaps	Check the model uses sensorimotor cortex, not artefacts

Table 3.1 — Methodology overview. Each stage is specified in detail in Sections 4 to 7.

3.1 Why subject-invariance is applied at the input, not only in the network

Most recent work attacks subject variability inside the network, through adversarial feature alignment or attention. Euclidean Alignment attacks it before the network sees the data: it whitens each subject's trials by their own mean spatial covariance, so every subject's data arrives with identity covariance. The operation is unsupervised — it needs only unlabelled trials from the new subject, never their labels — and costs one eigendecomposition of a 22×22 matrix, which is negligible against a forward pass.

This matters for deployment. A new user puts on the headset, records a minute of unlabelled motor imagery, and the alignment matrix is computed from it. No labelling, no supervised calibration session. That is what makes the 'without lengthy calibration' requirement operational rather than aspirational.

Euclidean Alignment — formal definition

For subject s with trials $X_1 \dots X_N$, each of shape C channels $\times$ T samples, compute the mean spatial covariance $R_s = (1/N) \sum_i X_i X_i^T$. Each trial is then aligned as $\tilde{X}_i = R_s^{-1/2} X_i$. After this transform the mean covariance of every subject equals the identity matrix, so the first-order distribution shift between subjects is removed. Reference: He and Wu, IEEE Transactions on Biomedical Engineering 67(2):399–410, 2020.

4. System Architecture — SAND-Net

SAND-Net (Subject-Agnostic Neural Decoder) is assembled from components taken deliberately from the reviewed literature, with one addition. The design rule was that every component must earn its place in the ablation; nothing is included because it appeared in a paper.

4.1 Component provenance and justification

Component	Taken from	Why it is included	Cost
Euclidean Alignment	He & Wu (2020)	Removes subject covariance shift; unsupervised, needs no target labels	≈0 parameters
Depthwise-separable conv block	EEGNet (Lawhern, 2018)	Learns CSP-like spatial filters at very low parameter cost	Low
Multi-scale dilated temporal conv	ATCNet (2023), CTNet (2024)	Mu and beta rhythms evolve on different timescales; one kernel size cannot capture both	Moderate
Windowed multi-head self-attention	EEG Conformer (2023), ATCNet (2023)	Long-range temporal dependency without the $O(T^2)$ cost and overfitting of full attention	Moderate
Supervised contrastive + domain-adversarial loss	Domain-generalisation literature	Pulls same-class trials from different subjects together in feature space	Training only; zero at inference
Class Activation Topography	EEG Conformer (2023)	Maps channel contribution per class for neurophysiological validation	Zero

Table 4.1 — Every component traces to a reviewed paper or to an explicit gap in them. The fifth row is the addition none of the four architectures has.

4.2 Layer specification

Configuration shown for BCI IV-2a: C = 22 channels, T = 875 samples (0.5–4.0 s post-cue at 250 Hz), 4 classes.

#	Layer	Configuration	Output shape
0	Euclidean Alignment	$R_s^{-1/2}$ applied per subject	22×875
1	Temporal convolution	F1 = 16, kernel (1, 64), padding same	$16 \times 22 \times 875$
2	Depthwise spatial conv	D = 2, kernel (22, 1), max-norm 1.0	$32 \times 1 \times 875$
3	BN → ELU → AvgPool(1,4) → Dropout 0.25	—	$32 \times 1 \times 218$
4	Separable convolution	F2 = 32, kernel (1, 16)	$32 \times 1 \times 218$
5	BN → ELU → AvgPool(1,8) → Dropout 0.25	—	$32 \times 1 \times 27$
6	Multi-scale temporal branch	3 parallel dilated convs, k = 4/8/16, dilation 1/2/4	96×27
7	Fusion	1×1 convolution	32×27
8	Windowed MHSA	5 overlapping windows, 2 heads, depth 2, learnable positional encoding	32×27
9	Global average pool → FC	32 → 4, softmax	4

Table 4.2 — SAND-Net layer specification. Target parameter budget is under 50,000, placing it between ShallowConvNet (47,364) and ATCNet (113,732).

4.3 Training objective

$$\mathbf{L}_{\text{total}} = \mathbf{L}_{\text{CE}} + \lambda_1 \cdot \mathbf{L}_{\text{SupCon}} + \lambda_2 \cdot \mathbf{L}_{\text{DANN}}$$

- $\mathbf{L}_{\text{CE}}$ — standard cross-entropy on the four motor imagery classes.

- $\mathbf{L}_{\text{SupCon}}$ — supervised contrastive loss computed across subjects: trials of the same class from different subjects are pulled together, trials of different classes are pushed apart. This is what forces the class structure to be subject-independent rather than subject-specific.
- $\mathbf{L}_{\text{DANN}}$ — a subject classifier attached through a gradient reversal layer. If the encoder's features can still identify who produced the trial, they carry subject information the decoder should not be relying on.

λ_1 and λ_2 are tuned on the validation subject only, never on the test subject. λ_2 is ramped from 0 following the standard DANN schedule to avoid destabilising early training.

5. Dataset Selection and Description

Three motor imagery datasets are used, in a tiered arrangement. All are public, all are standard benchmarks in this literature, and all can be downloaded programmatically through MOABB.

Dataset	Tier	Subjects	Channels	Classes	Rate	Trials/ subject
BCI Competition IV-2a	Primary	9	22	4	250 Hz	576
BCI Competition IV-2b	Secondary	9	3	2	250 Hz	720
High Gamma Dataset	Stretch	14	128	4	500 Hz	~880

Table 5.1 — Dataset tiers. The primary dataset carries the main claims; the secondary tests whether they transfer; the stretch tier is attempted only if the schedule allows and is not load-bearing for any conclusion.

5.1 Why these three

- BCI IV-2a is the common denominator of all four reviewed papers, so our reproduced baselines can be sanity-checked against published numbers before we trust anything else.
- BCI IV-2b uses only three electrodes (C3, Cz, C4), which sit inside 2a's montage. It tests whether the approach survives a drastic reduction in spatial information — directly relevant to low-cost consumer headsets.
- High Gamma Dataset has more subjects and higher-density coverage, giving a harder generalisation test if time permits.

5.2 Why SEED is excluded

The EEG Conformer paper also evaluates on SEED, and it was tempting to include it for breadth. SEED is an emotion recognition dataset: participants watch film clips and the task is to predict positive, neutral or negative affect. It contains no motor imagery and no movement labels. A model trained jointly on SEED and IV-2a would be predicting across two unrelated label spaces, which measures nothing about motor intent decoding. It is therefore excluded, and this is a deliberate decision rather than an oversight.

5.3 Preprocessing pipeline

1. Load raw recordings via MNE-Python / MOABB; retain the 22 EEG channels and discard the 3 EOG channels.
2. Band-pass filter 4–38 Hz using a 4th-order Butterworth filter applied forward and backward (zero-phase) to preserve temporal structure.
3. Epoch each trial from 0.5 s to 4.0 s after the cue, giving 875 samples per trial at 250 Hz.
4. Reject trials whose peak-to-peak amplitude exceeds 100 μV on any channel, after recording the rejection rate per subject.
5. Compute the Euclidean Alignment matrix per subject per session and apply it. For the held-out test subject this uses only their unlabelled trials.

- Standardise per channel using exponential moving standardisation, with statistics computed on the training subjects only.
- Augment the training split using Segmentation and Reconstruction — splitting trials into segments and recombining segments from same-class trials — following the procedure used by EEG Conformer and CTNet.

Leakage control

Normalisation statistics, augmentation and hyperparameter selection are derived exclusively from the training and validation subjects. The held-out test subject contributes only unlabelled trials to the alignment step, which is the same information a real deployment would have. Any deviation from this is a leak and invalidates the cross-subject claim.

6. Baseline Models

Five baselines are reproduced. Four are the architectures from the literature survey; the fifth is a classical method included because deep models are not automatically better on data of this size.

Baseline	Type	Parameters	Published IV-2a accuracy	Protocol used in source
FBCSP + LDA	Classical	—	~68%	Subject-specific
EEGNet	Compact CNN	2,548	68.67% / 71.50%	Varies by source
ShallowConvNet	CNN	47,364	66.41% / 67.48%	Varies by source
ATCNet	CNN + attention + TCN	113,732	81.10%	Train-Val-Test split
CTNet	CNN + transformer	Not stated	82.52%	Subject-specific
EEG Conformer	CNN + transformer	Not stated	78.66%	Hold-out

Table 6.1 — Published baselines. Parameter counts for EEGNet, ShallowConvNet and ATCNet are from the ATCNet reference implementation. Accuracies are as reported by the authors under their own protocols.

Reading Table 6.1 correctly

These numbers are not directly comparable. Each row uses a different train/test protocol, and some models appear with two different figures depending on which paper is reporting them. EEGNet is quoted at 68.67% by the ATCNet authors and 71.50% by the SATrans-Net authors — same model, same dataset, different evaluation. Re-running all of them under one protocol is precisely why Objective 2 exists.

Cross-subject figures are scarcer and lower. CTNet reports 58.64% on IV-2a under subject-independent evaluation; independent LOSO studies place EEGNet and Conformer around 68–70%, and recent work (SATrans-Net) reports 72.33%. Our target band is set from these, not from the within-subject column.

7. Experimental Plan and Evaluation Protocol

7.1 Primary protocol — leave-one-subject-out

For each of the 9 subjects in IV-2a, that subject is held out entirely as the test set. Of the remaining 8, one is held out as a validation subject for early stopping and hyperparameter selection, and 7 are used for training. This rotates 9 times and the reported figure is the mean and standard deviation across the 9 folds.

Using a held-out validation subject rather than a random split of the training subjects' trials is deliberate. Validating on trials from subjects that also appear in training would select hyperparameters that favour within-subject fit, which is the behaviour we are trying to avoid.

Fold	Training subjects	Validation subject	Test subject
1	S3 – S9 (7 subjects)	S2	S1
2	S1, S4 – S9	S3	S2
...	...	...	...
9	S2 – S7	S1	S9

Table 7.1 — LOSO fold structure. No trial from the test subject is ever seen with its label during training.

7.2 Secondary protocol — within-subject reference

Each model is also trained and tested within subject, session 1 to session 2, so our reproduced figures can be checked against the published ones. If our reproduced within-subject EEGNet lands far from the published value, the pipeline is wrong and the cross-subject numbers cannot be trusted either. This is a correctness check, not a headline result.

7.3 Calibration-efficiency experiment

This experiment tests H3 and produces the figure most directly tied to the clinical motivation. After LOSO training, the model is fine-tuned on n labelled trials from the held-out subject, for $n = 0, 5, 10, 20, 40$ and 80 , and accuracy is recorded at each point. Plotting accuracy against n for SAND-Net and each baseline shows how much labelled data a new user must supply before the system becomes usable. A model that reaches an acceptable threshold at $n = 10$ is far more deployable than one that needs $n = 80$, even if their $n = 80$ accuracies are identical.

7.4 Evaluation metrics

Metric	What it captures	Why included
Accuracy (mean $\pm$ std over 9 folds)	Overall correctness and its variability	Standard; the std across folds is as important as the mean
Cohen's kappa	Agreement corrected for chance	Standard in BCI; comparable to published work
Macro-F1	Per-class balance	Reveals a model that wins by ignoring a hard class
Confusion matrix	Which classes are confused	Feet vs tongue confusion is a known failure mode
Parameters, FLOPs, latency	Model cost	The 'lightweight' claim must be measured, not asserted

Metric	What it captures	Why included
Wilcoxon signed-rank test	Statistical significance across subjects	Paired, non-parametric, $n = 9$; the test EEG Conformer uses

Table 7.2 — Evaluation metrics. Significance is tested at $\alpha = 0.05$, paired across the 9 subjects.

Reporting only the mean accuracy would hide the behaviour that matters most. Cross-subject decoders frequently fail badly on one or two specific subjects — the so-called BCI-illiteracy phenomenon — and a model with a slightly lower mean but a much smaller standard deviation is the better assistive device. Per-subject results will therefore be reported in full, not only as an average.

8. Ablation Study Design

The ablation is the experiment that tests H1, and it is designed so that specific row comparisons answer specific questions.

#	EA	Multi-scale	Windowed attn	SupCon	DANN	What this row is
A1	—	—	—	—	—	EEGNet-equivalent baseline
A2	Yes	—	—	—	—	Alignment alone
A3	—	Yes	Yes	—	—	Capacity alone
A4	Yes	Yes	—	—	—	Alignment + multi-scale
A5	Yes	Yes	Yes	—	—	Full architecture, no DG losses
A6	Yes	Yes	Yes	Yes	—	Plus contrastive
A7	Yes	Yes	Yes	—	Yes	Plus adversarial
A8	Yes	Yes	Yes	Yes	Yes	Full SAND-Net

Table 8.1 — Ablation grid. Each configuration is run over all 9 LOSO folds.

8.1 Which comparison answers which question

- **A1 vs A2 vs A3** — the core test of H1. A2 isolates the gain from alignment with no added capacity; A3 isolates the gain from added capacity with no alignment. If $A2 - A1 > A3 - A1$, H1 is supported.
- **A3 vs A5** — does alignment still help once the architecture is already strong, or is its benefit absorbed by capacity?
- **A5 vs A6, A7, A8** — are the domain-generalisation losses complementary to input alignment, or redundant with it? This is the second half of H1.
- **A4 vs A5** — does windowed attention earn its parameters, or is the multi-scale convolution doing the work?

A separate run replaces windowed attention with full self-attention at matched depth, holding everything else fixed, to test H2.

9. Interpretability and Neurophysiological Validation

Interpretability here is not a presentation aid; it is a correctness check. EEG models are notorious for achieving high accuracy by learning eye-movement or muscle artefacts that happen to correlate with the cue, rather than the motor imagery itself. Two analyses are run.

9.1 Class Activation Topography

Following the method introduced with EEG Conformer, per-class channel contributions are projected onto a scalp topography. This yields one topographic map per motor imagery class, showing which electrodes drove the decision.

9.2 The falsification test

Motor imagery produces event-related desynchronisation contralateral to the imagined limb: imagining left-hand movement suppresses mu rhythm over the right sensorimotor cortex, near electrode C4, and right-hand imagery does the same over the left, near C3. This is established neurophysiology, independent of any model.

Pass / fail criterion

If the left-hand topography peaks near C4 and the right-hand topography peaks near C3, the model has learned genuine sensorimotor physiology. If the peaks are frontal, symmetric, or concentrated on peripheral electrodes, the model is likely exploiting ocular or muscular artefacts, and the accuracy figure is not measuring motor intent decoding regardless of how high it is.

This is a genuine test with a genuine failure mode, and the result will be reported either way.

9.3 Attention analysis

Attention weights across the five temporal windows are averaged per class and plotted against trial time. The expectation is that weight concentrates roughly 0.5 to 2.5 s after the cue, the interval in which event-related desynchronisation is strongest. Attention concentrated at trial onset would suggest the model is reading the cue artefact rather than the imagery.

10. Tools, Frameworks and Environment

Category	Choice	Role
Language	Python 3.10+	All implementation
Deep learning	PyTorch 2.x	Model definition, training, autograd for gradient reversal
EEG processing	MNE-Python	Loading, filtering, epoching, topographic plotting
Dataset access	MOABB	Programmatic download and standardised cross-subject splits
Reference models	Braindecode	Validated EEGNet, ShallowConvNet, Deep4Net implementations
Classical baseline	scikit-learn, pyRiemann	FBCSP + LDA
Numerics	NumPy, SciPy	Alignment matrices, filtering, statistical tests
Visualisation	Matplotlib, Seaborn, MNE topomaps	Results figures, topographies, confusion matrices

Category	Choice	Role
Experiment tracking	Weights & Biases or TensorBoard	Logging 230+ training runs with fold-level metrics
Compute	Google Colab (T4) / Kaggle (P100)	GPU training with checkpointing
Version control	Git + GitHub	Code, configs, seeds, and result tables

Table 10.1 — Software and hardware environment.

Using Braindecode's reference implementations rather than writing every baseline from scratch is a deliberate risk-reduction choice: a reproduction failure caused by our own reimplementing bug would be indistinguishable from a genuine result, and we do not have time in Phase 3 to debug four architectures.

10.1 Reproducibility measures

- Every run is seeded; seeds are recorded alongside results.
- Preprocessed tensors are cached to disk so that preprocessing is performed once, not once per model.
- Every experiment is defined by a YAML config committed to the repository, so any reported number can be regenerated by pointing the trainer at its config.
- Checkpoints are written every epoch to persistent storage, so a session timeout costs one epoch, not one fold.

11. Implementation Plan and Milestones

Phase 3 runs from 16 September to 3 October 2026 — 18 days. The plan is front-loaded: the pipeline and baselines must be working before the proposed model is written, because a proposed model evaluated on a broken pipeline is worthless.

Days	Dates	Milestone	Deliverable / exit criterion
1–2	16–17 Sep	Environment and data	All three datasets downloading; EDA plots produced
3–4	18–19 Sep	Preprocessing + Euclidean Alignment	Cached tensors; alignment verified (covariance $\approx$ identity)
5–7	20–22 Sep	Baselines, within-subject	EEGNet and Conformer reproduce published accuracy within ± 3 pts
8–10	23–25 Sep	LOSO harness + all baselines	9-fold LOSO results for all five baselines on IV-2a
11–13	26–28 Sep	SAND-Net implementation	Model trains end to end; parameter count under 50k confirmed
14–15	29–30 Sep	Ablation study	All 8 configurations $\times$ 9 folds complete
16–17	1–2 Oct	IV-2b, calibration curve, interpretability	Cross-dataset results; topographies; calibration figure
18	3 Oct	Consolidation	Final tables, figures and significance tests for the 2nd Review

Table 11.1 — Phase 3 schedule, aligned to the 2nd Review on 3 October 2026.

11.1 Compute budget

The schedule is only credible if the experiments fit the available GPU time. Estimated per-fold training times on a T4-class GPU:

Experiment set	Runs	Est. minutes per run	Est. total
Baselines, LOSO, IV-2a	5 models $\times$ 9 folds = 45	3 – 15	$\approx$ 6 h
SAND-Net, LOSO, IV-2a	9	8	$\approx$ 1.2 h
Ablation, 8 configs	8 $\times$ 9 = 72	8	$\approx$ 9.6 h
IV-2b, all models	6 $\times$ 9 = 54	3	$\approx$ 2.7 h
Calibration curve	6 points $\times$ 9 folds	2	$\approx$ 1.8 h
Contingency for reruns	—	—	$\approx$ 5 h
Total			$\approx$ 26 GPU-hours

Table 11.2 — Compute estimate. Kaggle provides roughly 30 GPU-hours per week, so this fits within the window provided preprocessing is cached and runs are checkpointed. Free-tier Colab alone would be tight and the ablation would need to be run over multiple sessions.

12. Risk Register

Risk	Likelihood	Impact	Mitigation
LOSO runtime exceeds available GPU hours	Medium	High	Cache preprocessed tensors; mixed-precision training; early stopping; drop the High Gamma tier first
Reproduced baselines miss published accuracy	Medium	High	Use Braindecode / official repositories; report reproduced and published figures side by side and explain the gap
Session timeouts lose training progress	High	Medium	Checkpoint every epoch to persistent storage; resumable trainer
SAND-Net does not beat the baselines	Medium	Medium	The ablation still answers H1; a negative result that shows capacity beats alignment is a valid and reportable finding
High Gamma Dataset too heavy to process	Medium	Low	Declared a stretch tier; no conclusion depends on it
Adversarial loss destabilises training	Medium	Medium	Ramp λ_2 from zero; A6 (contrastive only) is a working fallback configuration
Interpretability shows artefact reliance	Low	High	Report it honestly; tighten artefact rejection and re-run; this is the reason the check exists

Table 12.1 — Risk register with mitigations. Risks are listed with the response already decided, so Phase 3 does not lose days to deliberation.

13. Expected Outcomes and Success Criteria

Success is defined in advance so that the 2nd Review can be assessed against a fixed standard rather than a standard chosen after seeing the results.

Criterion	Target	Basis for the target
LOSO accuracy, IV-2a	$\geq 70\%$	Independent LOSO studies place EEGNet and Conformer at 68–70%; SATrans-Net reports 72.33%
Parameter count	$< 50,000$	Below ShallowConvNet (47,364) and well below ATCNet (113,732)
Std. dev. across subjects	Lower than the best baseline	Consistency matters more than peak accuracy for assistive use
Calibration efficiency	Within 5 pts of subject-specific accuracy using ≤ 20 labelled trials	Defines what 'without lengthy calibration' means in practice
Statistical significance	$p < 0.05$, Wilcoxon signed-rank	Same test used by EEG Conformer
Interpretability	Contralateral peaks at C3 / C4	Established event-related desynchronisation physiology

Table 13.1 — Pre-registered success criteria.

13.1 Deliverables at the 2nd Review (3 October 2026)

- Working implementation of the full pipeline, five baselines and SAND-Net, in a public repository.
- LOSO results table for all models on IV-2a and IV-2b, with per-subject breakdown and significance tests.
- Completed ablation table with the H1 verdict stated explicitly.
- Calibration-efficiency curve.
- Class Activation Topography figures with the neurophysiological validation result.

13.2 Statement on novelty

The 0th Review answered 'Partially' to the question of sufficient novelty, and recorded that further investigation was needed. That investigation is now complete, and the position has changed. The contribution is no longer described as a combination of existing techniques. It is: (i) the first evaluation of EEGNet, ATCNet, CTNet and EEG Conformer under a single identical leave-one-subject-out protocol, which the published literature does not provide; (ii) a controlled test of whether input-space subject-invariance or architectural capacity contributes more to cross-subject decoding; and (iii) a calibration-efficiency characterisation that states how much labelled data a new user actually needs. Points (ii) and (iii) are questions the reviewed literature does not answer, and both produce a useful result whichever way they resolve.

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